

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460885.908 +/- 0.023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.033
Transit depth (Rp/Rs)^2	2.45 +/- 1.04 [%]
Orbital Inclination (inc)	86.38 +/- 1.67
Airmass coefficient 1 (a1)	1.007 +/- 0.013
Airmass coefficient 2 (a2)	-0.004 +/- 0.09
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	#2 - [269, 314]
Optimal Aperture	4.04
Optimal Annulus	9.23
Transit Duration (day)	0.065 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.033 vs 0.14195069566578392 (deviation 0.46σ)
Duration fit vs published	0.065 d vs 0.0875 d (deviation 0.68σ)
Mid-time O–C	-0.4 min
Depth SNR	4.8
Residual scatter	1.25 %

Light curve

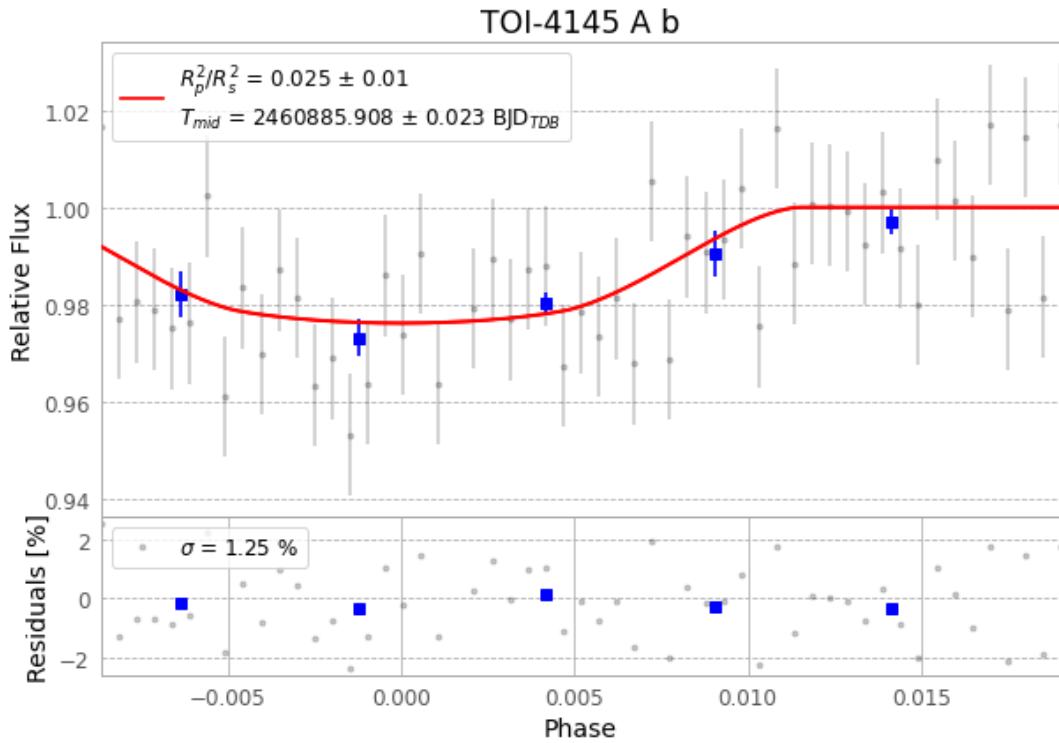


Figure 1 — FinalLightCurve_TOI-4145 A b_2025-07-29.png (SHA-256 6addf294097d88bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 17551019fc843c49...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

55 FITS frames (36 MB), TOI4145250729085715.FITS → TOI4145250729113917.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4145-250729.log (d55bdecb0de2...), FinalLightCurve_TOI-4145 A b_2025-07-29.png (6addf294097d...), FinalLightCurve_TOI-4145 A b_2025-07-29.csv (710779cd0815...)

Compute resources

Wall time 92.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:03Z.